

MODEL ANSWERS

Q1(a) [2 Marks]

Bloom: BL 1–2 (Remember / Understand) | CO1

Answer the following:

- (a) Write the expressions for the radial and transverse components of acceleration in polar coordinates (a_r and a_θ) and identify the centripetal term. [0.5]

Ans:

In polar coordinates the acceleration components are:

$$a_r = \ddot{r} - r\dot{\theta}^2 \text{ (radial component)}$$

$$a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} \text{ (transverse component)}$$

The term $-r\dot{\theta}^2$ in a_r is the centripetal acceleration, directed radially inward. The term $2\dot{r}\dot{\theta}$ in a_θ is the Coriolis component arising from simultaneous radial and angular motion.

- (b) State Hamilton's principle (principle of least action) and explain its relationship to the Euler-Lagrange equation. [0.5]

Ans:

Hamilton's principle states that the actual path taken by a mechanical system between two time instants t_1 and t_2 is the one that makes the action integral stationary:

$$\delta S = \delta \int_{(t_1 \text{ to } t_2)} L \, dt = 0, \text{ where } L = T - V.$$

Applying the calculus of variations to this condition yields the Euler-Lagrange equation:

$$d/dt(\partial L / \partial \dot{q}) - \partial L / \partial q = 0.$$

- (c) Differentiate between holonomic and non-holonomic constraints. Give one example of each from a mechanical system. [0.5]

Ans:

Holonomic constraint: can be expressed as $f(q_1, q_2, \dots, t) = 0$ (depends only on coordinates and time, not velocities).

Example: A bead on a wire — its position is restricted to the wire curve.

Non-holonomic constraint: involves velocities or inequalities and cannot be reduced to a coordinate-only equation.

Example: A wheel rolling without slipping — the constraint $dx = R \, d\theta$ relates velocity components but cannot be integrated into a position-only equation for all paths.

- (d) State the parallel axis theorem for moment of inertia and give its mathematical expression. [0.5]

Ans:

The parallel axis theorem states that the moment of inertia about any axis parallel to one through the center of mass is:

$$I = I_{cm} + md^2$$

where I_{cm} = moment of inertia about the CM axis, m = total mass, d = perpendicular distance between the two axes. I is always smallest about the CM axis.

Q1(b) [2 Marks]

Bloom: BL 2–3 (Understand / Apply) | CO1, CO2

- (a) A particle moves along a spiral with $r = 3 \, \text{m}$, $\dot{r} = 0$, and $\dot{\theta} = 2 \, \text{rad/s}$ (constant). Calculate the centripetal acceleration and state its direction. [0.5]

Ans:

For circular motion with $r = 3 \, \text{m}$, $\dot{r} = 0$, $\dot{\theta} = 2 \, \text{rad/s}$:

$$a_{\text{centripetal}} = r\dot{\theta}^2 = 3 \times 2^2 = 3 \times 4 = 12 \, \text{m/s}^2$$

$$\text{(In polar form: } a_r = \ddot{r} - r\dot{\theta}^2 = 0 - 12 = -12 \, \text{m/s}^2)$$

Direction: radially inward, toward the center of curvature.

- (b) Explain the concept of generalized coordinates. For a double pendulum in a plane, state the number of DOF and identify the generalized coordinates. [0.5]

Ans:

Generalized coordinates are the minimum set of independent coordinates needed to completely describe a system's configuration. They need not be Cartesian — they can be angles, arc lengths, etc.

A double pendulum in a plane has 2 DOF.

Generalized coordinates: θ_1 (angle of first link) and θ_2 (angle of second link), both measured from the vertical.

(c) Define the Coriolis acceleration. State the expression for its magnitude and explain in which physical situations it becomes significant. [0.5]

Ans:

The Coriolis acceleration arises when an object moves with velocity v_{rel} in a reference frame rotating at angular velocity Ω :

$$a_{Coriolis} = 2\Omega \times v_{rel}, \text{ magnitude} = 2\Omega v_{rel}$$

It is perpendicular to both Ω and v_{rel} . It becomes significant in large-scale meteorology (weather systems), long-range ballistics, and fast-rotating machinery where the product Ωv_{rel} is comparable to other acceleration terms.

(d) A uniform rod of mass $m = 3 \text{ kg}$ and length $L = 1.2 \text{ m}$ is pivoted at one end. Calculate its moment of inertia about the pivot using the parallel axis theorem. [0.5]

Ans:

$$\text{For a uniform rod: } I_{cm} = (1/12)mL^2 = (1/12) \times 3 \times 1.2^2 = (1/12) \times 3 \times 1.44 = 0.36 \text{ kg}\cdot\text{m}^2.$$

$$\text{Parallel axis theorem: } I_{pivot} = I_{cm} + m(L/2)^2$$

$$= 0.36 + 3 \times (0.6)^2 = 0.36 + 3 \times 0.36 = 0.36 + 1.08$$

$$= 1.44 \text{ kg}\cdot\text{m}^2.$$

$$(\text{Check: } I_{pivot} = (1/3)mL^2 = (1/3) \times 3 \times 1.44 = 1.44 \text{ kg}\cdot\text{m}^2 \quad \square)$$

Solve any TWO from Q2, Q3, Q4 [3 Marks each = 6 Marks]

Q2 [3 Marks]

Bloom: BL 3 (Apply) | CO1, CO2

A simple pendulum consists of a point mass $m = 2 \text{ kg}$ attached to a massless, inextensible string of length $L = 0.5 \text{ m}$. The bob swings in a vertical plane under gravity ($g = 9.81 \text{ m/s}^2$). Using the angle θ from the vertical as the generalized coordinate, analyze the system using the Lagrangian method. Use the following relations: Lagrangian $L = T - V$; Euler-Lagrange equation: $d/dt(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$; for small angles: $\sin \theta \approx \theta$.

(a) State the number of degrees of freedom of this system and identify the generalized coordinate. [0.5]

Ans:

The pendulum has 1 degree of freedom (DOF = 1).

Generalized coordinate: θ — the angle of the string measured from the vertical.

(b) Write the position of the mass in Cartesian coordinates (x, y) in terms of θ and L . [0.5]

Ans:

Taking the pivot as the origin, with x horizontal and y vertically downward:

$$x = L \sin \theta = 0.5 \sin \theta$$

$$y = L \cos \theta = 0.5 \cos \theta$$

$$\text{Differentiating: } \dot{x} = L\dot{\theta} \cos \theta, \dot{y} = -L\dot{\theta} \sin \theta$$

(c) Derive the kinetic energy T in terms of θ , $\dot{\theta}$, m , and L . [0.5]

Ans:

$$v^2 = \dot{x}^2 + \dot{y}^2 = L^2\dot{\theta}^2 \cos^2 \theta + L^2\dot{\theta}^2 \sin^2 \theta = L^2\dot{\theta}^2$$

$$T = \frac{1}{2}mv^2 = \frac{1}{2} \times 2.0 \times 0.5^2 \times \dot{\theta}^2 = \frac{1}{2}mL^2\dot{\theta}^2$$

$$T = \frac{1}{2}(2.0)(0.5^2)\dot{\theta}^2 = 0.25 \dot{\theta}^2 \text{ J}$$

(d) Write the potential energy V in terms of θ , m , g , and L . [0.5]

Ans:

Taking the pivot as the datum (y positive downward):

$$V = -mgy = -mgL \cos \theta = -2.0 \times 9.81 \times 0.5 \times \cos \theta$$

$$V = -9.81 \cos \theta \text{ J}$$

(e) Form the Lagrangian $L = T - V$ and apply the Euler-Lagrange equation to derive the equation of motion. [0.5]

Ans:

$$L = T - V = \frac{1}{2}mL^2\dot{\theta}^2 + mgL \cos \theta$$

$$\text{Euler-Lagrange: } d/dt(\partial L/\partial \dot{\theta}) - \partial L/\partial \theta = 0$$

$$\partial L/\partial \dot{\theta} = mL^2\dot{\theta} = 2.0 \times 0.5^2 \times \dot{\theta} = 0.5 \dot{\theta}$$

$$d/dt(mL^2\dot{\theta}) = mL^2\ddot{\theta} = 0.5 \ddot{\theta}$$

$$\partial L/\partial \theta = -mgL \sin \theta = -9.81 \sin \theta$$

$$\text{EOM: } 0.5 \ddot{\theta} + 9.81 \sin \theta = 0$$

$$\text{Simplifying (divide by } mL^2): \ddot{\theta} + (g/L) \sin \theta = 0$$

$$\rightarrow \ddot{\theta} + (9.81/0.5) \sin \theta = 0 \rightarrow \ddot{\theta} + 19.62 \sin \theta = 0$$

- (f) For small oscillations ($\sin \theta \approx \theta$), determine the natural frequency of the pendulum. [0.5]

Ans:

For small angles, $\sin \theta \approx \theta \rightarrow \ddot{\theta} + (g/L)\theta = 0$

This is SHM with $\omega_n^2 = g/L = 9.81/0.5 = 19.62$

$\omega_n = \sqrt{19.62} = 4.429 \text{ rad/s}$

Time period $T = 2\pi/\omega_n = 2\pi/4.429 = 1.419 \text{ s}$

Q3 [3 Marks]

Bloom: BL 3–4 (Apply / Analyze) | CO2

A horizontal turntable rotates at a constant angular velocity $\Omega = 5 \text{ rad/s}$ about a vertical axis. A small ball at $r = 2 \text{ m}$ from the center is pushed radially outward at a constant relative speed $v_{\text{rel}} = 3 \text{ m/s}$ ($\dot{r} = 3 \text{ m/s}$, $\ddot{r} = 0$). The angular acceleration of the turntable is zero ($\alpha = 0$).

- (a) Write the general expression relating the acceleration observed from the fixed (inertial) frame to quantities measured in the rotating frame. Identify each term. [0.5]

Ans:

The fixed-frame (inertial) acceleration, expressed in terms of rotating-frame quantities, is:

$$\mathbf{a}_{\text{fixed}} = \mathbf{a}_{\text{rel}} + 2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}} + \boldsymbol{\alpha} \times \mathbf{r} + \boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$$

where:

$\mathbf{a}_{\text{rel}}$ = acceleration of the object in the rotating frame ($\ddot{r} = 0$ here),

$2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}}$ = Coriolis acceleration,

$\boldsymbol{\alpha} \times \mathbf{r}$ = Euler (angular acceleration) term (zero here since $\alpha = 0$),

$\boldsymbol{\Omega} \times (\boldsymbol{\Omega} \times \mathbf{r})$ = centripetal acceleration (directed radially inward).

- (b) Calculate the centripetal acceleration magnitude at $r = 2 \text{ m}$. [0.5]

Ans:

$$a_{\text{centripetal}} = \Omega^2 r = 5.0^2 \times 2.0 = 25 \times 2.0 = 50.0 \text{ m/s}^2$$

Direction: radially inward (toward the axis of rotation).

- (c) Calculate the Coriolis acceleration magnitude. [0.5]

Ans:

$$a_{\text{Coriolis}} = 2\Omega v_{\text{rel}} = 2 \times 5.0 \times 3.0 = 30.0 \text{ m/s}^2$$

- (d) State the direction of the Coriolis acceleration relative to the radial velocity. [0.5]

Ans:

By the right-hand rule for $2\boldsymbol{\Omega} \times \mathbf{v}_{\text{rel}}$ ($\boldsymbol{\Omega}$ along vertical, $\mathbf{v}_{\text{rel}}$ radially outward): the Coriolis acceleration is perpendicular to $\mathbf{v}_{\text{rel}}$, directed tangentially in the direction opposite to the platform's rotation.

- (e) Calculate the total acceleration magnitude of the object as observed from the fixed (inertial) frame. [0.5]

Ans:

$$|a_{\text{total}}| = \sqrt{a_{\text{cent}}^2 + a_{\text{Cor}}^2} = \sqrt{50.0^2 + 30.0^2} \\ = \sqrt{2500 + 900} = \sqrt{3400} = 58.31 \text{ m/s}^2$$

- (f) If the ball continues at the same radial speed, find the centripetal acceleration when $r = 4 \text{ m}$ and comment on how it changes. [0.5]

Ans:

$$\text{At } r = 4 \text{ m: } a_{\text{centripetal}} = \Omega^2 r = 5.0^2 \times 4 = 100.0 \text{ m/s}^2$$

Compared to 50.0 m/s^2 at $r = 2.0 \text{ m}$, the centripetal acceleration has doubled. It increases linearly with radial distance ($a_{\text{cent}} \propto r$), while the Coriolis acceleration (30.0 m/s^2) remains unchanged since v_{rel} is constant.

Q4 [3 Marks]

Bloom: BL 4–5 (Analyze / Evaluate) | CO2, CO3

A uniform rod of mass $m = 2 \text{ kg}$ and length $L = 1.2 \text{ m}$ is pivoted at one end and oscillates as a physical (compound) pendulum under gravity. Take $g = 9.81 \text{ m/s}^2$. Use: $I_{\text{rod cm}} = (1/12)mL^2$; parallel axis theorem: $I = I_{\text{cm}} + md^2$; period of physical pendulum: $T = 2\pi\sqrt{I_{\text{pivot}} / (mgd)}$, where d = distance from pivot to center of mass.

- (a) Calculate the moment of inertia of the rod about its center of mass (I_{cm}). [0.5]

Ans:

$$I_{\text{cm}} = (1/12)mL^2 = (1/12) \times 2.0 \times 1.2^2 \\ = (1/12) \times 2.0 \times 1.44 = 0.2400 \text{ kg}\cdot\text{m}^2$$

- (b) Using the parallel axis theorem, calculate the moment of inertia about the pivot at one end (I_{pivot}). [0.5]

Ans:

$$I_{\text{pivot}} = I_{\text{cm}} + m(L/2)^2 = 0.2400 + 2.0 \times (1.2/2)^2 \\ = 0.2400 + 2.0 \times 0.6^2 = 0.2400 + 0.7200 \\ = 0.9600 \text{ kg}\cdot\text{m}^2$$

(Check: $I_{\text{pivot}} = (1/3)mL^2 = (1/3) \times 2.0 \times 1.44 = 0.9600 \text{ kg}\cdot\text{m}^2$ □)

- (c) Find the distance d of the center of mass from the pivot. [0.5]

Ans:

$$d = L/2 = 1.2/2 = 0.6 \text{ m}$$

(The center of mass of a uniform rod is at its midpoint.)

- (d) Write the equation of motion for small-angle oscillations and determine the natural frequency ω_n . [0.5]

Ans:

Restoring torque: $\tau = -mgd \sin \theta \approx -mgd\theta$ (small angle)

Equation of motion: $I_{\text{pivot}} \ddot{\theta} + mgd \theta = 0$

$$\rightarrow \ddot{\theta} + (mgd / I_{\text{pivot}}) \theta = 0$$

$$\omega_n = \sqrt{(mgd / I_{\text{pivot}})} = \sqrt{(2.0 \times 9.81 \times 0.6 / 0.9600)}$$

$$= \sqrt{(11.7720 / 0.9600)} = \sqrt{12.2625}$$

$$= 3.5018 \text{ rad/s} \approx 3.502 \text{ rad/s}$$

- (e) Calculate the time period of oscillation T . [0.5]

Ans:

$$T = 2\pi / \omega_n = 2\pi / 3.5018 = 1.7943 \text{ s} \approx 1.794 \text{ s}$$

- (f) Compare this period with that of a simple pendulum of the same length ($L = 1.2 \text{ m}$) and explain the difference. [0.5]

Ans:

Simple pendulum of same length $L = 1.2 \text{ m}$:

$$T_{\text{simple}} = 2\pi\sqrt{(L/g)} = 2\pi\sqrt{(1.2/9.81)} = 2\pi \times 0.3497$$

$$= 2.1975 \text{ s} \approx 2.198 \text{ s}$$

Physical pendulum period (1.794 s) < Simple pendulum period (2.198 s).

This is because the physical pendulum's mass is distributed along its length, with some mass closer to the pivot than L . The effective pendulum length $L_{\text{eff}} = I_{\text{pivot}}/(md) = 0.9600/(2.0 \times 0.6) = 0.8000 \text{ m} < 1.2 \text{ m}$.

— End of Set 2 Model Answers —